

# LAB EXPERIMENT RECORD

## Set 8: Employee Performance Analysis

|                           |                                           |
|---------------------------|-------------------------------------------|
| Department                | CSE-Ds                                    |
| Course Code / Course Name | DSA0601 / Data Handling and Visualization |
| Experiment                | Data Visualization using R                |
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| Register Number           | 192473018                                 |
| Date                      | 03 October 2026                           |
| Day                       | Saturday                                  |

### AIM

To analyze employee performance data using R by creating a line chart, bar chart, and table, and to develop an interactive dashboard for exploring employee performance.

### DATASET: Employee Performance

| Employee ID | Department | Years of Service | Performance Score |
|-------------|------------|------------------|-------------------|
| 1           | Sales      | 5                | 85                |
| 2           | HR         | 3                | 92                |
| 3           | Marketing  | 7                | 78                |

### PROGRAM 1: Line Chart for Employee Performance

#### Aim

To create a line chart to visualize the performance trend of employees based on their years of service.

#### Program

R

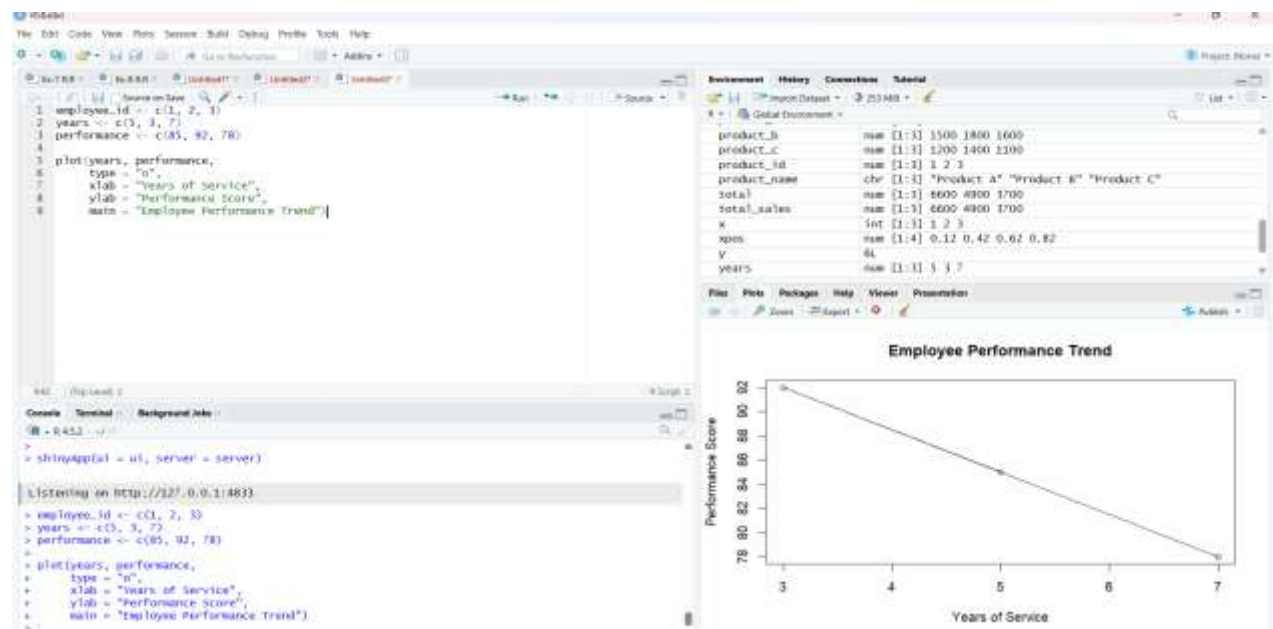
```

employee_id <- c(1, 2, 3)
years <- c(5, 3, 7)
performance <- c(85, 92, 78)

plot(years, performance,
     type = "o",
     xlab = "Years of Service",
     ylab = "Performance Score",
     main = "Employee Performance Trend")

```

## Output



A line chart showing the performance scores of employees according to their years of service.

## Observation

The employee with 3 years of service has the highest performance score of 92, while the employee with 7 years of service has the lowest score of 78.

## PROGRAM 2: Bar Chart for Department Distribution

### Aim

To create a bar chart showing the distribution of employees across different departments.

### Program

R

```

department <- c("Sales", "HR", "Marketing")
count <- c(1, 1, 1)

barplot(count,

        names.arg = department,

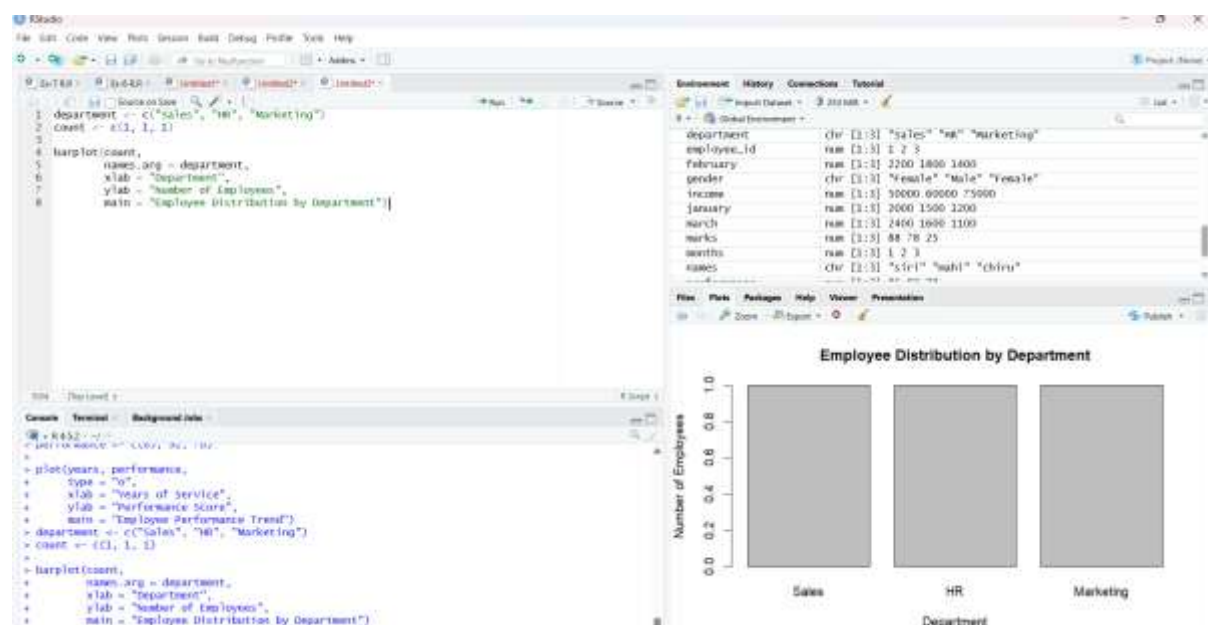
        xlab = "Department",

        ylab = "Number of Employees",

        main = "Employee Distribution by Department")

```

## Output



A bar chart displaying the number of employees in Sales, HR, and Marketing departments.

## Observation

Each department has one employee. Therefore, all three departments have equal employee distribution.

## PROGRAM 3: Employee Performance Data Table

### Aim

To create a table displaying the performance information of each employee.

### Program

R

```
employee_id <- c(1, 2, 3)
```

```
department <- c("Sales", "HR", "Marketing")
```

```
years <- c(5, 3, 7)
```

```
performance <- c(85, 92, 78)
```

```
employee_data <- data.frame(
```

```
  Employee_ID = employee_id,
```

```
  Department = department,
```

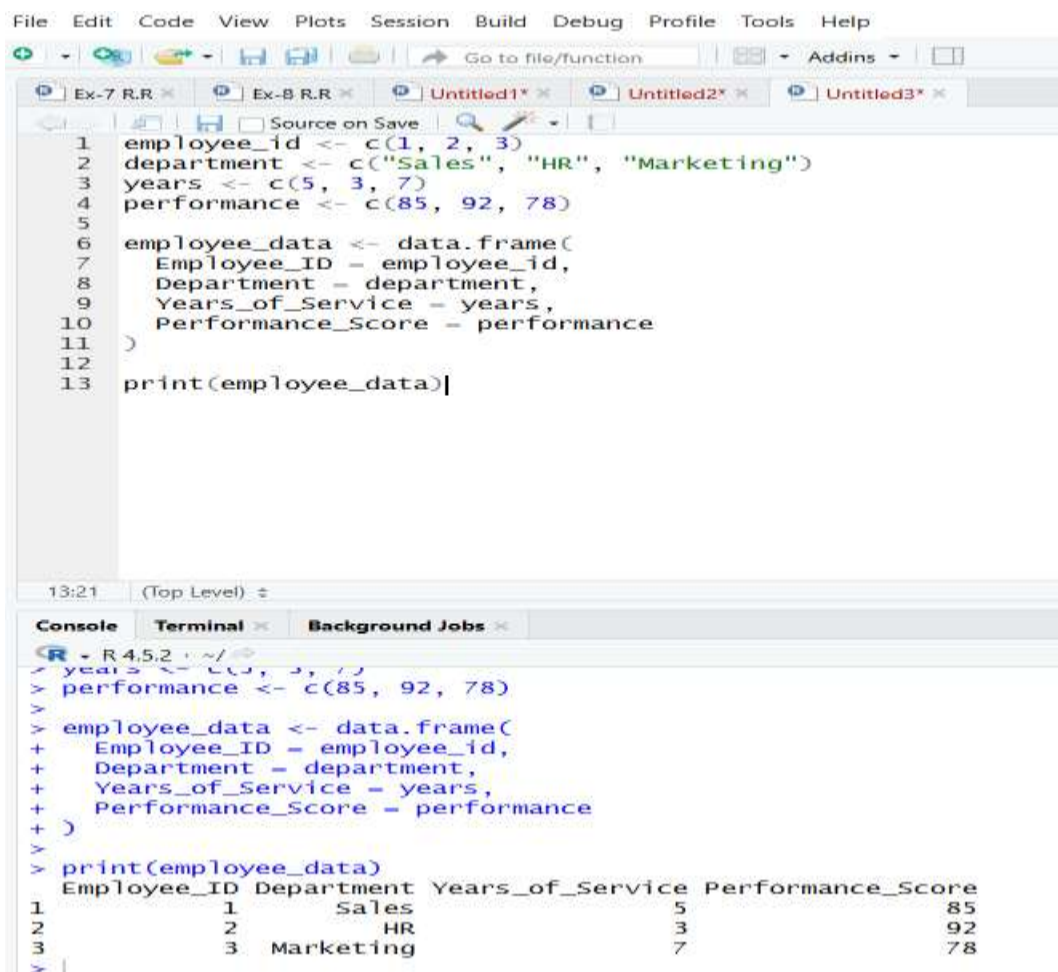
```
  Years_of_Service = years,
```

```
  Performance_Score = performance
```

```
)
```

```
print(employee_data)
```

## Output



The screenshot shows the RStudio IDE with the following code in the editor:

```
1 employee_id <- c(1, 2, 3)
2 department <- c("Sales", "HR", "Marketing")
3 years <- c(5, 3, 7)
4 performance <- c(85, 92, 78)
5
6 employee_data <- data.frame(
7   Employee_ID = employee_id,
8   Department = department,
9   Years_of_Service = years,
10  Performance_Score = performance
11 )
12
13 print(employee_data)
```

The console output shows the execution of the code and the resulting data frame:

```
> years <- c(5, 3, 7)
> performance <- c(85, 92, 78)
>
> employee_data <- data.frame(
+   Employee_ID = employee_id,
+   Department = department,
+   Years_of_Service = years,
+   Performance_Score = performance
+ )
>
> print(employee_data)
  Employee_ID Department Years_of_Service Performance_Score
1           1     Sales                5                85
2           2        HR                3                92
3           3 Marketing                7                78
```

The table displays Employee ID, Department, Years of Service, and Performance Score.

### **Observation**

The employee performance information is displayed in a structured tabular format.

## **PROGRAM 4: Interactive Employee Performance Dashboard**

### **Aim**

To develop an interactive dashboard combining the line chart, bar chart, and employee performance table.

### **Program**

R

```
library(shiny)

employee_id <- c(1, 2, 3)

department <- c("Sales", "HR", "Marketing")

years <- c(5, 3, 7)

performance <- c(85, 92, 78)

employee_data <- data.frame(
  Employee_ID = employee_id,
  Department = department,
  Years_of_Service = years,
  Performance_Score = performance
)

ui <- fluidPage(
  titlePanel("Employee Performance Analysis Dashboard"),
  mainPanel(
    plotOutput("lineChart"),
    plotOutput("barChart"),
    tableOutput("employeeTable")
  )
)
```

```

)

server <- function(input, output) {

  output$lineChart <- renderPlot({

    plot(employee_data$Years_of_Service,

         employee_data$Performance_Score,

         type = "o",

         xlab = "Years of Service",

         ylab = "Performance Score",

         main = "Employee Performance Trend")

  })

  output$barChart <- renderPlot({

    count <- table(employee_data$Department)

    barplot(count,

            xlab = "Department",

            ylab = "Number of Employees",

            main = "Employee Distribution by Department")

  })

  output$employeeTable <- renderTable({

    employee_data

  })

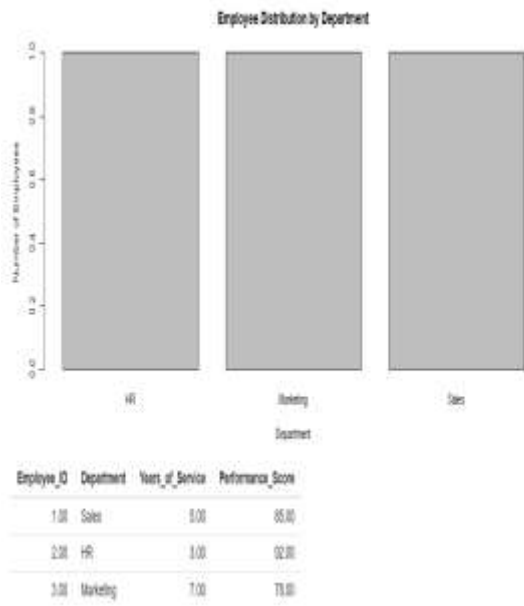
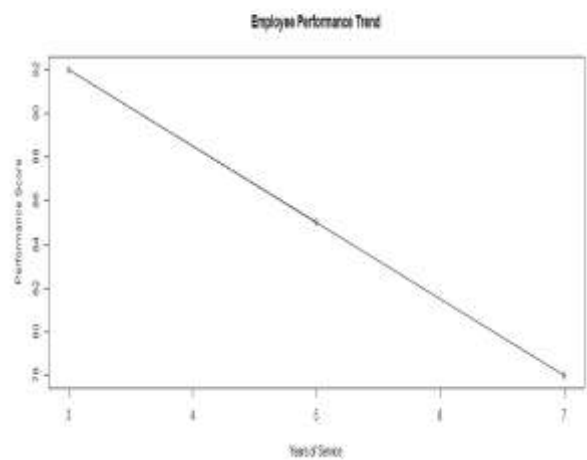
}

shinyApp(ui = ui, server = server)

```

## Output

# Employee Performance Analysis Dashboard



## Observation

The dashboard provides a simple way to view and analyze employee performance data using charts and a table in one place.